

Candidate Name : _____

CT Group : _____

Index no : _____

PIONEER JUNIOR COLLEGE
JC 2 Preliminary Examination



MATHEMATICS
Higher 1 (Syllabus 8863)

Tuesday

9 SEPTEMBER 2008

Additional material:

- Answer paper
- Formula booklet

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your full name, index number and CT group on all the work you hand in.

Write in dark blue or black pen on both sides of the papers.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

Attach this question paper with your answers, and arrange your answers in numerical order.

For Examiner's Use			
Qn	Marks	Qn	Marks
1		8	
2		9	
3		10	
4		11	
5		12	
6		13	
7		Total	

This question paper consists of 7 printed pages and 1 blank page.

Section A: Pure Mathematics (40 marks)

- 1** Solve the simultaneous equations:

$$\log_4(3x-10) - \log_4(y+1) = 1,$$

$$\frac{8^x}{2^{y^2}} = 4.$$

[4]

- 2** Find the set of values of x , in exact form, for which

$$3x^2 - x - 2 \geq 0.$$

[2]

Hence, deduce the solution of

$$3x^2 - |x| - 2 \geq 0.$$

[2]

- 3** On the same axes, for $0 \leq x \leq 2\pi$, sketch the graph of $y = \sin x$ and the graph of $y = 2 \sin 3x$.

[3]

Hence,

- (i)** state the period of $y = 2 \sin 3x$,

[1]

- (ii)** state the number of solutions to the equation

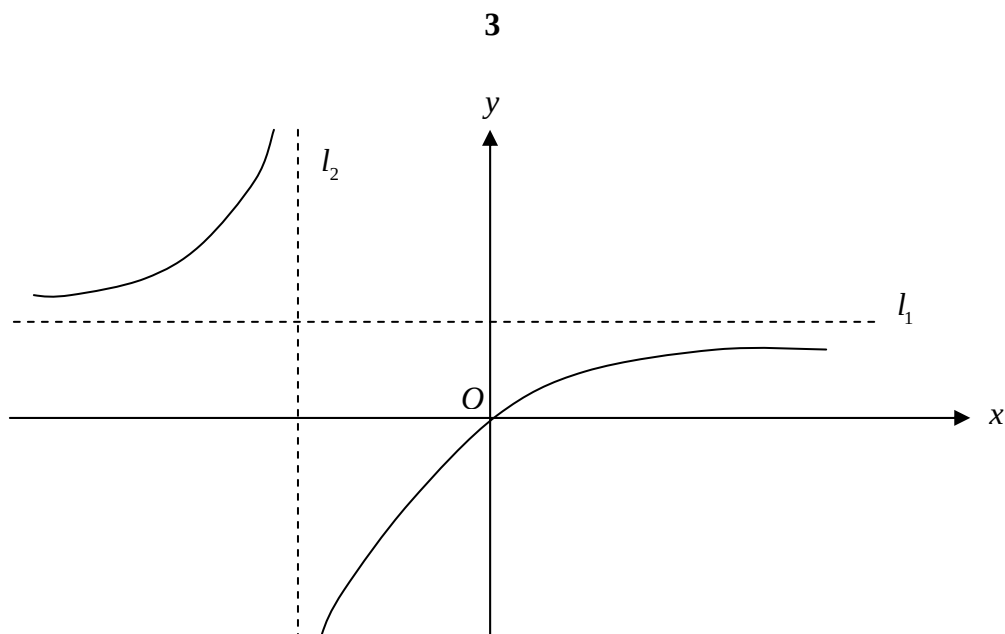
$$2 \sin 3x - \sin x = 0 \quad \text{for } 0 \leq x \leq 2\pi,$$

[1]

- (iii)** deduce the number of solutions to the equation

$$|2 \sin 3x| - \sin x = 0 \quad \text{for } 0 \leq x \leq 2\pi.$$

[1]



The diagram shows a sketch of the curve $y = \frac{6x}{x+3}$. The lines l_1 and l_2 are asymptotes to the curve. State the equations of the lines l_1 and l_2 . [3]

(i) State the range of $y = \frac{6x}{x+3}$. [1]

(ii) Find the area of the finite region bounded by the curve, the x -axis, and the line $x = 6$. Give your answer in exact form. [4]

5 The functions f and g are defined as follows:

$$\begin{aligned} f : x &\mapsto e^x, & x &\in \square, \\ g : x &\mapsto x^2 - 3x - 10, & x &\in \square. \end{aligned}$$

(i) Find the range of f and hence give a reason why gf exists. [2]

(ii) Find the exact value of x such that $gf(x) = 0$. [3]

(iii) The function g has an inverse if its domain is restricted to $x \geq k$. State the least value of k for g^{-1} to exist. [1]

(iv) By expressing g in the form $(x+a)^2 + b$, or otherwise, find an expression for g^{-1} corresponding to the domain found in (iii). [3]

- 6 (a) A curve has equation $y = 4x + \frac{5}{x^2}$. Using differentiation, show that the x -coordinate of the turning point on the curve is $\sqrt[3]{\frac{5}{2}}$. Determine the nature of this point. [4]
- (b) The height of an upright cone is twice the radius, r , of its circular base. It is known that the volume of the cone is changing at the rate of $15 \text{ cm}^3 \text{ min}^{-1}$ when the radius is 3 cm. Find, in exact form, the rate of increase of the radius of the cone at this instant, and hence find the rate of increase of the base area of the cone at the same instant.

[The volume of a cone is given by $\frac{1}{3} \times (\text{base area}) \times (\text{height})$] [5]

Section B: Statistics (60 marks)

- 7 A random variable X follows a binomial distribution such that $X \sim B(n, p)$. Given that the mean of X is 2.6 and the variance of X is 2.08, find n and p . Hence calculate $P(|X - 3| < 2)$. [4]
- 8 There are 800 new students joining PJC and the student council would like to conduct a survey on their perception of the college.
- (i) Describe how the student council would choose a systematic random sample of 50 new students. [3]
- (ii) State briefly one advantage and one disadvantage of this sampling method. [2]
- 9 On Friday, Saturday and Sunday evenings, Jack will either stay at home to do his tutorials, or go out to watch a movie. However, he will not do both of the two activities on the same evening on these three days. The probability that he does his tutorials on Friday evenings is p . On Saturday and Sunday evenings, the probability that he does his tutorials given that he did his tutorials on the previous evening is $\frac{1}{3}$, and the probability that he does his tutorials given that he went for a movie on the previous evening is $\frac{2}{5}$.
- (i) Draw a tree diagram to illustrate the above given information. [3]
- It is given that the probability he does his tutorials on Sunday evening is $\frac{589}{1575}$.
- (ii) Show that $p = \frac{1}{7}$. [2]
- (iii) Find the probability that he does his tutorials on Friday evening, given that he does his tutorials on Sunday evening. Give your answer as a fraction in its simplest form. [3]

- 10** Onions are packed in bags which the farmer claims contain 1.75 kg each, on average. A random sample of 100 bags is examined and the mass, x kg, of the contents of each bag is determined. It is found that $\sum x = 189.1$ and $\sum x^2 = 401.74$.
- (i)** Calculate an unbiased estimate of the population mean and population variance of the mass of the contents of a bag, giving your answers to 3 decimal places. **[2]**
 - (ii)** Test, at the 3% significance level, whether the data indicate that the mean mass of the contents of a bag differs from 1.75 kg. **[4]**
 - (iii)** If the test had been for an increase in the mean at the 3% significance level, what would have been the result of the test? Justify your answer. **[2]**
 - (iv)** State, giving a reason, whether it is necessary for the mass of the contents of a bag to have a normal distribution for the test to be valid. **[1]**
- 11** Gloves manufactured by a certain factory are packed in boxes of 50. On average, 2% of the gloves will be defective. A box which contains 3 or more defective gloves will be rejected. Show that the probability that a given box will be rejected is 0.078428. **[2]**
- (i)** The boxes are transferred from the factory to the warehouse using a fork-lift which carries 10 boxes per trip. In a particular week, the fork-lift made 100 trips. Find the probability that the mean number of rejected boxes per trip is at least 1 and at most 3, giving your answer to 3 significant figures. **[4]**
 - (ii)** There are 500 boxes stored in the warehouse. Using a suitable approximation, find the probability that the number of rejected boxes is at most 30. **[4]**

- 12** Orange juice is sold in bottles of two sizes, standard and large. The content, in litres, of a randomly chosen standard bottle follows a normal distribution with mean 0.760 and standard deviation 0.008. The content, in litres, of a randomly chosen large bottle follows a normal distribution with mean 1.010 and standard deviation 0.009.
- (i)** Two large bottles are chosen at random. Find the probability that one of the large bottles contains less than 1 litre of orange juice and the other contains more than 1 litre of orange juice. **[2]**
 - (ii)** Find the probability that a box of 10 randomly chosen standard bottles contains at least three bottles, each with less than 0.750 litres. **[4]**
 - (iii)** Find the probability that the orange juice in four randomly chosen standard bottles is more than three times the orange juice in a randomly chosen large bottle. **[4]**
- 13** A study was carried out in 8 countries to measure the average monthly household income, x , in US\$1000 against y , the infant mortality per 1000 live births. The data is recorded in the table below:

x	5.11	6.17	4.71	4.15	3.87	2.72	10.13	5.44
y	4.23	3.98	4.81	6.30	7.82	9.45	11.71	4.15

- (i)** Calculate the product moment correlation coefficient, and the equation of the estimated least squares regression line of y on x in the form $y = a + bx$. **[3]**
- (ii)** Sketch the scatter diagram of the data, as shown on your GC. Giving a reason, identify a data pair which should be regarded as suspect. **[3]**
- (iii)** Remove the suspect data pair, and calculate the product moment correlation coefficient, and the least squares regression line of y on x , for the revised data. **[2]**
- (iv)** Comment on your answers to parts (i) and (iii). **[2]**

Two other countries A and B , are later surveyed on their respective average monthly household income. The value of x for A is to be 1.2, and the value of x for B is to be 4.95. For each of these two countries, find the estimated value of y , using the least squares regression line obtained in part (iii), and state, with a reason in each case, whether it would be appropriate to regard this value as a valid prediction. **[4]**

[End of Paper]

Pure Maths (40 marks) Solutions:

1.

$$\log_4(3x-10) - \log_4(y+1) = 1$$

$$\log_4 \frac{(3x-10)}{(y+1)} = 1$$

$$\frac{(3x-10)}{(y+1)} = 4$$

$$3x-10 = 4y+4$$

$$\Rightarrow 3x = 4y+14 \text{-----}(A)$$

$$\frac{8^x}{2^{y^2}} = 4$$

$$\frac{2^{3x}}{2^{y^2}} = 2^2$$

$$2^{3x-y^2} = 2^2$$

$$3x - y^2 = 2$$

$$\Rightarrow 3x = 2 + y^2 \text{-----}(B)$$

Solving equations (A) and (B),

We have

$$4y+14 = 2+y^2$$

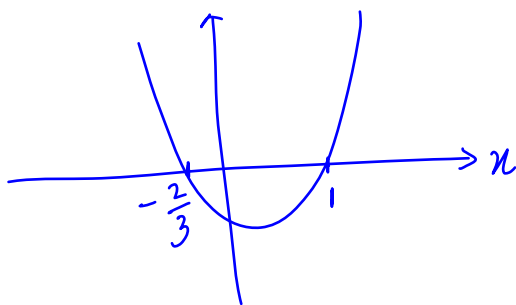
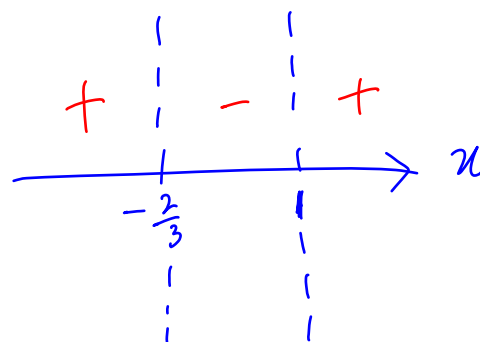
$$y^2 - 4y - 12 = 0$$

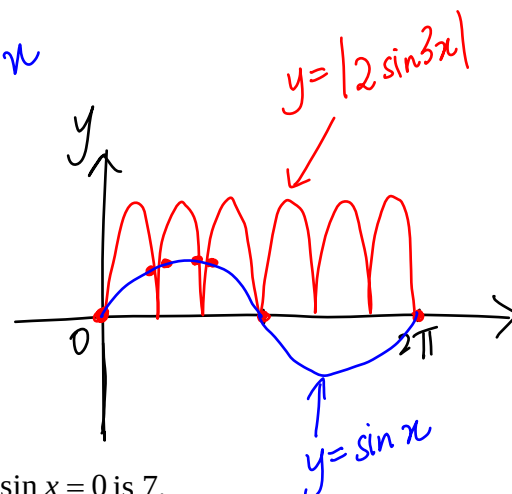
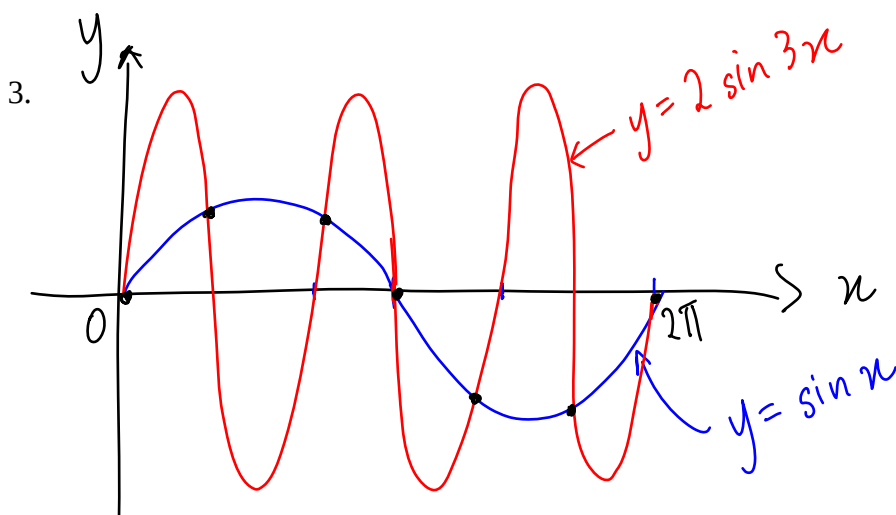
$$(y-6)(y+2) = 0$$

$$\therefore y = 6 \text{ or } y = -2(\text{NA})$$

$$\text{Thus, } x = \frac{2+(6)^2}{3} = \frac{38}{3} \approx 12.7$$

2. $3x^2 - x - 2 \geq 0$

ORHence, solution is $x \leq -\frac{2}{3}$ or $x \geq 1$.Replace x by $|x|$.Thus solution to $3x^2 - |x| - 2 \geq 0$ is $|x| \leq -\frac{2}{3}$ (NA) or $|x| \geq 1$.Thus, $x \geq 1$ or $x \leq -1$.



- (i) Period of $y = 2 \sin 3x$ is $\frac{2\pi}{3}$.
- (ii) From graph, number of solutions to the equation $2 \sin 3x - \sin x = 0$ is 7.
- (iii) Number of solutions to the equation $|2 \sin 3x| - \sin x = 0$ is 7.

4.

$$\begin{array}{r} 6 \\ x+3 \overline{) 6x} \\ \underline{-(6x+18)} \\ -18 \end{array}$$

Thus $\frac{6x}{x+3} = 6 - \frac{18}{x+3}$.

Thus $l_1 : y = 6$ and $l_2 : x = -3$.

(i) $R_f = \mathbb{R} \setminus \{6\}$

(ii) Required area =

$$\begin{aligned} & \int_0^6 6 - \frac{18}{x+3} dx \\ &= \left[6x - 18 \ln|x+3| \right]_0^6 \\ &= [6(6) - 18 \ln|6+3|] - [6(0) - 18 \ln|0+3|] \\ &= 36 - 18 \ln 9 + 18 \ln 3 \\ &= 36 - 18 \ln 3^2 + 18 \ln 3 \\ &= 36 - 36 \ln 3 + 18 \ln 3 \\ &= 36 - 18 \ln 3 \end{aligned}$$

5.

- (i) $R_f = (0, \infty)$
 Since $R_f = (0, \infty) \subseteq \mathbb{R} = D_g$
 Thus gf exists.

- (ii) $gf(x)$
 $= (e^x)^2 - 3e^x - 10$
 $= e^{2x} - 3e^x - 10$
 $= (e^x - 5)(e^x + 2)$
 $gf(x) = 0$
 $(e^x - 5)(e^x + 2) = 0$
 $\Rightarrow e^x = 5 \text{ or } e^x = -2 \text{ (NA)}$
 Thus, $e^x = 5 \Rightarrow x = \ln 5$

- (iii) Least value of $k = \frac{3}{2}$.

- (iv) $x^2 - 3x - 10 = (x - \frac{3}{2})^2 - \frac{9}{4} - 10$
 $= (x - \frac{3}{2})^2 - \frac{49}{4}$

$$\text{Let } y = (x - \frac{3}{2})^2 - \frac{49}{4}$$

$$y + \frac{49}{4} = (x - \frac{3}{2})^2$$

$$x = \frac{3}{2} \pm \sqrt{(y + \frac{49}{4})}$$

$$\text{Since } x \geq \frac{3}{2}, \quad g^{-1} = \frac{3}{2} + \sqrt{(x + \frac{49}{4})}.$$

Alternative: (Using quadratic formula)

or

$$\text{Let } y = x^2 - 3x - 10$$

$$x^2 - 3x - (10 + y) = 0$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(1)[-(10 + y)]}}{2(1)}$$

$$x = \frac{3 \pm \sqrt{9 + 4(10 + y)}}{2}$$

$$x = \frac{3 \pm \sqrt{49 + 4y}}{2}$$

$$x = \frac{3 \pm \sqrt{49 + 4y}}{2}$$

$$x = \frac{3}{2} + \frac{\sqrt{49 + 4y}}{2} \text{ or } x = \frac{3}{2} - \frac{\sqrt{49 + 4y}}{2}$$

6. (a) $y = 4x + \frac{5}{x^2}$.

$$\frac{dy}{dx} = 4 - \frac{10}{x^3}$$

$$\frac{dy}{dx} = 0 \Rightarrow 4 - \frac{10}{x^3} = 0 \Rightarrow 4 = \frac{10}{x^3} \Rightarrow x^3 = \frac{5}{2} \Rightarrow x = \sqrt[3]{\frac{5}{2}}$$

Using 2nd derivative test,

$$\frac{d^2y}{dx^2} = \frac{30}{x^4} \text{ is +ve when } x = \sqrt[3]{\frac{5}{2}}.$$

Thus it is a minimum point.

Using 1st derivative test,

x	$\left(\sqrt[3]{\frac{5}{2}}\right)^{-}$	$\sqrt[3]{\frac{5}{2}}$	$\left(\sqrt[3]{\frac{5}{2}}\right)^{+}$
Sign of $\frac{dy}{dx}$	-ve	0	+ve
Outline of curve			

Thus, it is a minimum point.

$$\begin{aligned} \text{(b) Volume of cone, } V &= \frac{1}{3}\pi r^2(2r) = \frac{2\pi r^3}{3} \\ \Rightarrow \frac{dV}{dr} &= 2\pi r^2 \end{aligned}$$

At $r = 3$,

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$15 = 2\pi(3)^2 \times \frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{5}{6\pi}$$

Base Area of cone, $A = \pi r^2$

$$\frac{dA}{dr} = 2\pi r$$

At $r = 3$,

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$= 2\pi(3) \left(\frac{5}{6\pi} \right)$$

$$= 5\text{cm}^2 \text{min}^{-1}$$

Statistics (60 marks) Solutions:

$$np = 2.6 \text{ \& } np(1-p) = 2.08 \Rightarrow 2.6(1-p) = 2.08 \Rightarrow p = 0.2 \text{ \& } n = 13$$

$$7 \quad \therefore X \sim B(13, 0.2)$$

$$\Rightarrow P(|X - 3| < 2) = P(-2 < X - 3 < 2) = P(1 < X < 5) = P(2 \leq X \leq 4) = 0.667$$

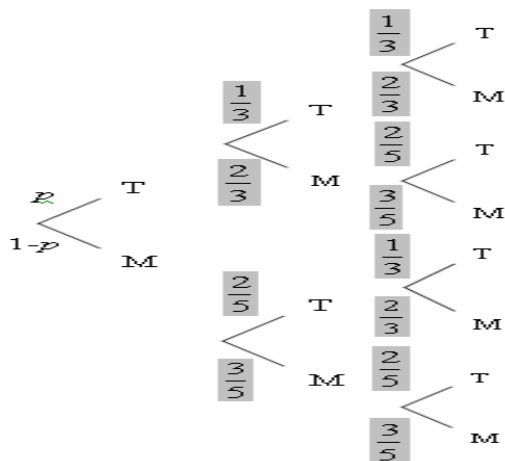
$$8 \quad (i) \quad 800/50 = 16$$

To obtain a sample by systematic sampling, we randomly choose a random number in the range of 1 to 16, say 9.

Then we choose the 9th, 25th, 41th, ..., 793th students from the college register for the sample.

- (ii) Advantage: Quick & easy to conduct as compared to other types of sampling.
Disadvantage: Could have an inappropriate number of male & female students in the population sample.

$$9 \quad \text{Fri} \quad \text{Sat} \quad \text{Sun}$$



$$(ii) \quad p \left(\frac{1}{3} \right) \left(\frac{1}{3} \right) + p \left(\frac{2}{3} \right) \left(\frac{2}{5} \right) + (1-p) \left(\frac{2}{5} \right) \left(\frac{1}{3} \right) + (1-p) \left(\frac{3}{5} \right) \left(\frac{2}{5} \right) = \frac{589}{1575}$$

$$\text{Solving, } p = \frac{1}{7}$$

- (iii) Let F be the event Jack does his tutorial on Friday and S be the event Jack does his tutorial on Sunday

$$\text{Therefore probability} = \frac{P(F \cap S)}{P(S)} = \frac{\left(\frac{1}{7} \right) \left(\frac{1}{3} \right) \left(\frac{1}{3} \right) + \left(\frac{1}{7} \right) \left(\frac{2}{3} \right) \left(\frac{2}{5} \right)}{\frac{589}{1575}} = \frac{85}{589}$$

$$10 \text{ (i)} \quad \sum x = 189.1 \Rightarrow \bar{x} = \frac{189.1}{100} = 1.891 \text{ (exact)}$$

$$\Rightarrow \sum x^2 = 401.74 \Rightarrow s^2 = \frac{1}{100-1} \left[401.74 - \frac{(189.1)^2}{100} \right] = 0.44598 \approx 0.446 \text{ (3 dp)}$$

$$(ii) \quad H_0 : \mu = 1.75 \text{ vs } H_1 : \mu \neq 1.75$$

Level of significance: 3%

$$\text{Standardised test statistic: } z = \frac{\bar{x} - \text{"claimed value"}}{s/\sqrt{n}} = \frac{1.891 - 1.75}{\sqrt{\frac{0.44598}{100}}} = 2.1114 \approx 2.11$$

From GC, $p\text{-value} = 0.0347 (> 0.03)$

Since the $p\text{-value}$ is more than the level of significance, hence we do not reject H_0 . There is insufficient evidence, at the 3% level, to indicate that the mean mass of the contents of a bag differs from 1.75 kg.

$$(iii) \quad \text{If test at } H_0 : \mu = 1.75 \text{ vs } H_1 : \mu > 1.75$$

From GC, $p\text{-value} = 0.0174 (< 0.03)$

Since the $p\text{-value}$ is less than the level of significance, we reject H_0 . There is sufficient evidence, at the 3% level, to indicate that the mean mass of the contents of a bag has increased from 1.75 kg.

$$(iv) \quad \text{Not necessary to assume that the mass of the contents of a bag has a normal Distribution since sample size } n \text{ is large, by CLT,}$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ approximately.}$$

$$11 \quad \text{Let } X \text{ represent the number of defective gloves out of 50}$$

$$X \sim B(50, 0.02)$$

$$P(X \geq 3) = 1 - P(X \leq 2) = 0.078428$$

$$(i) \quad \text{Let } Y \text{ represent the number of rejected boxes per trip}$$

$$Y \sim B(10, 0.078428) \Rightarrow E(Y) = 0.78428 \text{ \& Var}(Y) = 0.72277$$

By CLT, $\bar{Y} \sim N(0.78428, 0.0072277)$ approximately

$$P(1 \leq \bar{Y} \leq 3) = 0.00558$$

$$(ii) \quad W \sim B(500, 0.078428)$$

$$n=500 \text{ is large, } np=39.214 > 5 \text{ \& } n(1-p)=460 > 5 \text{ Var}(W) = 36.139$$

$$\Rightarrow W \sim N(39.214, 36.139) \text{ approximately}$$

$$P(W \leq 30) \Rightarrow P(W < 30.5)(c.c.) = 0.0736$$

- 12 Let $S \square$ amount of orange juice in a standard bottle in litres $\Rightarrow S \square N(0.76, 0.008^2)$
 & $L \square$ amount of orange juice in a large bottle in litres $\Rightarrow L \square N(1.01, 0.009^2)$

(i) $2 \times P(L < 1) \times P(L > 1)$

$$= 2 \times \text{normcdf}(-10^{99}, 1, 1.010, 0.009) \times \text{normcdf}(1, 10^{99}, 1.010, 0.009) \\ = 0.231$$

(ii) $P(S < 0.75) = \text{normcdf}(-10^{99}, 0.75, 0.76, 0.008) = 0.10565$

Let X be the no. of standard bottles which contains < 0.75 l orange juice out of 10 bottles

$$\therefore X \square B(10, 0.10565) \Rightarrow P(X \geq 3) = 1 - P(X \leq 2) = 1 - \text{binomcdf}(10, 0.10565, 2) = 0.0803$$

(iii) $E(S_1 + S_2 + S_3 + S_4 - 3L) = 4 \times 0.76 - 3 \times 1.01 = 0.01$

$$\text{Var}(S_1 + S_2 + S_3 + S_4 - 3L) = 4 \times 0.008^2 + 3^2 \times 0.009^2 = 0.000985$$

$$P(S_1 + S_2 + S_3 + S_4 > 3L)$$

$$= P(S_1 + S_2 + S_3 + S_4 - 3L > 0)$$

$$= \text{normcdf}\left(0, 10^{99}, 0.01, \sqrt{0.000985}\right)$$

$$= 0.625$$

- 13 Using GC to obtain $r = 0.332$
 Using GC to obtain $y = 4.29 + 0.429x$
 Sketch using GC to show 7 points near the line with negative gradient and 1 outlier
 State “(10.13, 11.71) is the outlier”
 State “Outlier” or “far from the rest” or equivalent
 Using GC to obtain $r = -0.948$
 Obtain from GC: $y = 14.0 - 1.78x$
 State “Correlation now strongly negative” or equivalent (increased household income associated with reduced mortality rate)
 State “Regression line now completely different” or equivalent
 Substitute $x = 1.2$ & $x = 4.95$ in regression line (iii)
 Obtain $y_A = 11.9$ & $y_B = 5.19$
 A: extrapolation – not appropriate
 B: interpolation – appropriate